

Soinik Ghosh

PhD Researcher, Biomedical Engineering • Co-Founder & CTO, Hridae Medical Technologies

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Research Profile

I develop interpretable and reliable clinical tools from brain and heart signals and medical images, and build automated imaging pipelines and a local AI agent for research and clinical workflows. My PhD builds EEG-based screening and triage tools and intelligent clinical decision-support systems for neurology, with a focus on dementia and epilepsy, designed so that clinicians can see what drives each result and so that performance holds on patients from new hospitals. First-author work on reduced-electrode EEG for cognitive screening, with neurologists at IMS BHU, was accepted at **GlobalSouthAI @ NeurIPS 2026**. As Co-Founder and CTO of Hridae Medical Technologies, I lead the development of wearable monitoring for precision cardiology and neurology. I studied physiology and neuroscience before moving into engineering.

Education

Indian Institute of Technology (BHU) Varanasi — *Varanasi, India*
PhD in Biomedical Engineering (CGPA: 9.6/10) — 2023 – Present
Thesis area: EEG-based screening, triage and clinical decision support.

University of Calcutta — *Kolkata, India*
M.Sc in Physiology (Neuroscience specialisation), First Class (CGPA: 7.4/10) — 2019 – 2021
Thesis: "Neuropsychophysiological Behaviour of Adenosine: Crosstalks between Neurotransmitters"

Surendranath College, University of Calcutta — *Kolkata, India*
B.Sc in Human Physiology (Honours) — 2016 – 2019

Positions

Co-Founder & CTO, Hridae Medical Technologies Pvt. Ltd. — *2026 – Present*
Lead technology development for wearable monitoring in precision cardiology and neurology: devices, analysis models and clinician review software, developed with clinicians.

Doctoral Researcher & Teaching Assistant, School of Biomedical Engineering, IIT (BHU) — *2023 – Present*
EEG-based screening and triage tools and intelligent clinical decision-support systems, designed to be interpretable and reliable. In collaboration with the Department of Neurology, IMS BHU. Experimental design, data collection and analysis, algorithm development, scientific writing and teaching.

Publications

Published and accepted work only. Full list on Google Scholar.

Workshop and conference papers

- **S. Ghosh**, R. N. Chaurasia, S. Sharma, N. Sharma. *Decomposing a clinical EEG montage for low-channel cognitive triage: placement, age referencing and output design.* GlobalSouthAI Workshop @ NeurIPS 2026, Sydney (accepted). [code]
- **S. Ghosh**, C. Parikshith, F. Yasmeen, A. K. Srivastava, S. Sharma, N. Sharma. *Real-time brain signals monitoring system using single-channel EEG for BCI applications.* IEEE AKGEC 2024.
- P. Chavakula, **S. Ghosh**, N. Sharma, D. Kumar. *Development of a haptic exo-suit with tactile feedback in hand rehabilitation.* IEEE INSPECT 2024.
- *A hybrid classical-quantum model for enhanced MRI-based brain tumor classification using transfer learning and quantum neural networks.* IEEE ICCSAI 2025 (co-author).

Journal articles

- **S. Ghosh**, S. Sharma, N. Sharma. *GenEEG: Improving epileptic EEG detection through patient-adaptive latent diffusion and continual learning.* Computers in Biology and Medicine 201, 111398 (2025).

Book chapters

- **S. Ghosh**, F. Yasmeen, S. Sharma, N. Sharma. *Decoding the neural dynamics of motor imagery and execution: an EEG-based investigation into conscious hand movement representation.* In: Cognitive Biomarkers and BCI Applications, Proceedings of MBCC 2025, Springer, 2026, pp. 311–322.
- F. Yasmeen, **S. Ghosh**, P. Jangra, N. Sharma, S. Sharma. *ME-GSE: an end-to-end classification of motor imagery EEG signals using graph-based spectral embedding.* In: Cognitive Biomarkers and BCI Applications, Proceedings of MBCC 2025, Springer, 2026.

Featured Research

Low-channel EEG for cognitive triage *GlobalSouthAI @ NeurIPS 2026 • IIT (BHU), IMS BHU*

Studied how far a clinical EEG can be simplified for cognitive screening, using memory-clinic recordings and independent cohorts from other hospitals. A few well-placed electrodes screened about as well as the full clinical montage, and EEG added value beyond age alone; detecting mild cognitive impairment remains an open problem. Proposes a simplified recording design for prospective device studies.

Current Research

- **Low-channel EEG screening for dementia:** electrode placement, sampling and output design for cognitive triage devices, validated at hospitals unseen in training.
- **Epilepsy monitoring and seizure forecasting:** patient-adaptive EEG monitoring and probabilistic forecasting of seizure risk.
- **Wearable monitoring (Hridae):** wearable devices and analysis software for precision cardiology and neurology.
- **Automated medical image processing:** automated pipelines that prepare and analyse images across medical imaging modalities.
- **Local AI agent:** on-device AI agent running on an 8 GB laptop GPU, trained to carry out coding, research and complex clinical workflow tasks.
- **Offline cognitive assessment:** browser-based battery across eleven cognitive domains with on-device measures and within-person change tracking.

Selected Projects

GenEEG: patient-adaptive synthetic EEG *IIT (BHU)*

Latent diffusion for synthetic seizure EEG with continual learning one patient at a time. [code]

Hridayam: AI-assisted non-invasive cardiovascular imaging *Smart India Hackathon 2024, Winner*

Multi-sensor point-of-care cardiovascular screening. [code]

Firefly: VR-guided robotic exosuit for stroke rehabilitation *PRAGATI Hackathon 2023, 2nd runner-up*

Immersive VR environment coupled with a robotic exoskeleton. [code]

Motor-imagery BCI for prosthetic control *IIT (BHU)*

Real-time translation of motor-imagery EEG into prosthetic arm commands; live decoding on a Raspberry Pi 5.

GemSAM: agentic, explainable multimodal medical image analysis on edge *Kaggle MedGemma Impact Challenge 2026*

MedGemma and SAM 2 pipeline built for edge hardware. [code]

Interictal network dynamics *IIT (BHU), IMS BHU*

Interictal epileptiform discharges as dynamic network events via source connectivity and graph neural networks. [code]

CogniDhi and HippoSphere AI *Stanford Longevity Design Challenge 2025–26; MIT Media Lab Moodeng AI Challenge*

Cognitive health platform; lightweight computer vision for animal behaviour monitoring. [CogniDhi] [HippoSphere]

Skills

Signals & models	EEG, ECG and PCG processing; qEEG and spectral features; ICA artefact handling; source connectivity; GNNs; latent diffusion; continual and patient-adaptive learning; medical image processing; LLM-based agents
Evaluation	Subject-grouped cross-validation; external-cohort testing; shortcut and confound checks; paired bootstrap and permutation tests; calibration
Engineering	Python, PyTorch, MNE, MATLAB, C/C++, C#, TypeScript, React, GCP; embedded EEG, low-noise PCB design, Raspberry Pi and microcontrollers; Unity VR/XR
Clinical & lab	Clinical study design and ethics applications; clinician review workflows; cell culture, molecular biology, animal studies

Honours & Training

- **Winner**, Smart India Hackathon 2024 (Hardware Edition), Government of India.
- **2nd Runner-up**, PRAGATI Hackathon 2023, iHub Drishti (TIH), IIT Jodhpur.
- **GATE 2023**, Life Sciences, All India Rank 1313.
- g.tec BCI & Neurotechnology Spring School 2025: invasive and non-invasive BCI.
- Neurogati / CNS Lab, IIT Madras: computational brain modelling and brain-inspired AI.
- SERB Karyashala, IIT Roorkee: AI in human brain-computer interaction.
- CSIR-IICB Skill Development Programme: molecular cloning, protein expression and characterisation.

Beyond Science

Independent artist, musician and filmmaker (music, IMDb).